

bbTips 2.0

How a BBCode becomes a pop-up — the reference-and-enrich model

A standalone, provider-based extension that registers game BBcodes with phpBB's s9e TextFormatter and emits nothing but a *reference* anchor — `data-wowhead="item=50468"` — which WowHead's official `tooltips.js` enriches into a live pop-up entirely in the browser.

Package avathar/bbtips v2.0.0-rc2 · type `phpbb-extension`

Requires phpBB 3.3 (declared $\geq 3.3.0$; effectively $\geq 3.3.11$ — see §5) · PHP ≥ 8.1

Providers World of Warcraft · Diablo 4 (GW2 planned)

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1 Executive summary

bbTips 2.0 is a ground-up rewrite of the phpBB 3.0 MOD as a standalone phpBB 3.3 extension. A reader gets the same thing they always did — WowHead-style tooltips that pop up over in-line game links — but how that data reaches them is turned inside out.

Three decisions shape the codebase, and the rest of this document is mostly their consequences.

1. **Games are providers.** The core handles the BBCode plumbing, runtime loading, ACP and public API; a *provider* (World of Warcraft, Diablo 4, GW2 later) supplies one game's tags, link format and client runtime. Providers are tagged services gathered into a registry, so tag registration, the editor toolbar and runtime de-duplication all discover a new one on their own. Enablement isn't fully data-driven yet — provider config, the ACP page and the migration still name each game explicitly — and §6 walks through exactly what a third game would touch.
2. **The server ships a reference, not a tooltip.** A tag renders to a bare anchor that carries an entity *reference* and nothing else (`<a data-wowhead="item=50468&domain=classic">`). No scraping, no cache table, no API keys — the MOD's entire server-side fetch-and-store layer is gone, not ported.
3. **The browser does the enrichment.** WowHead's `tooltips.js`, loaded once per page, reads those references and fills them in: it swaps the link text for the live name, adds the icon and quality color as the page settles, and draws the full stat card on hover. All of it is cached in the reader's browser.

Something real was lost here: name and text tags. The 1.0.7 MOD let you write the entity's *name* between the brackets — `[spell]Power Word: Shield[/spell]`, `[item]Ardent Guard[/item]`, `[craft]Recipe: Vial of the Sands[/craft]`. bbTips 2.0 takes a **numeric ID only** (`[spell]17[/spell]`). So authors now have to look the ID up first, and — the part that stings — **every name-based tag already sitting in old posts stops producing a tooltip** the moment you upgrade; it falls back to plain text or a dead link. `renameLinks` will put the live name back once there's a valid ID behind it, but it can't turn a name that was never an ID into one.

This wasn't a choice so much as a consequence. Name lookups only ever worked because the old MOD **scraped WowHead's search pages by name** (`simple_html_dom.php`, `wowhead_patterns.php`) and cached what it found. That road is closed now for two independent reasons. WowHead's modern `tooltips.js` resolves **by numeric ID only** — the name-search endpoint the MOD relied on no longer exists. And scraping WowHead server-side was **against their Terms of Service** to begin with, and brittle besides: the 1.x changelog reads like a running battle against their redesigns. Tearing out that scrape-and-cache layer was the point of the rewrite, and name resolution lived entirely inside it. Keeping the feature would have meant keeping the thing we set out to remove. Numeric-only is the price of a data path that's legitimate and needs no babysitting.

2 Generation 1 → Generation 2

What a reader sees hasn't changed. Nearly everything behind it has. The MOD leaned on a fragile server-side scraper; 2.0 emits a thin reference and lets an official browser-side runtime do the rest.

BBTIPS 1.X — PHPBB 3.0 MOD

- **Interception:** legacy template *display hook* rewriting message text at view time.
- **Resolution:** server scrapes WowHead XML/JSON by name or ID, synchronously, at render.
- **Storage:** a DB cache table of resolved objects (`dbal.php`).
- **Output:** server builds tooltip-ready markup from HTML "pattern" files.
- **Client:** ships WowHead's `power.js` locally.
- **Structure:** procedural includes; one class per object type.
- **Extending:** edit core parser + add classes.

BBTIPS 2.0 — PHPBB 3.3 EXTENSION

- **Interception:** `s9e\TextFormatter` *configurator event* registers real BBCodes.
- **Resolution:** none server-side — the browser resolves references via WowHead's API.
- **Storage:** none — no tooltip data ever touches phpBB.
- **Output:** an XSL template emits a bare reference anchor.
- **Client:** loads WowHead's official `tooltips.js` from the Zam CDN.
- **Structure:** namespaced DI services, PSR-style; core services unit-tested (PHPUnit + CI).
- **Extending:** a provider class + service tag covers discovery; enablement config/ACP/migration still edited per game (\$6).

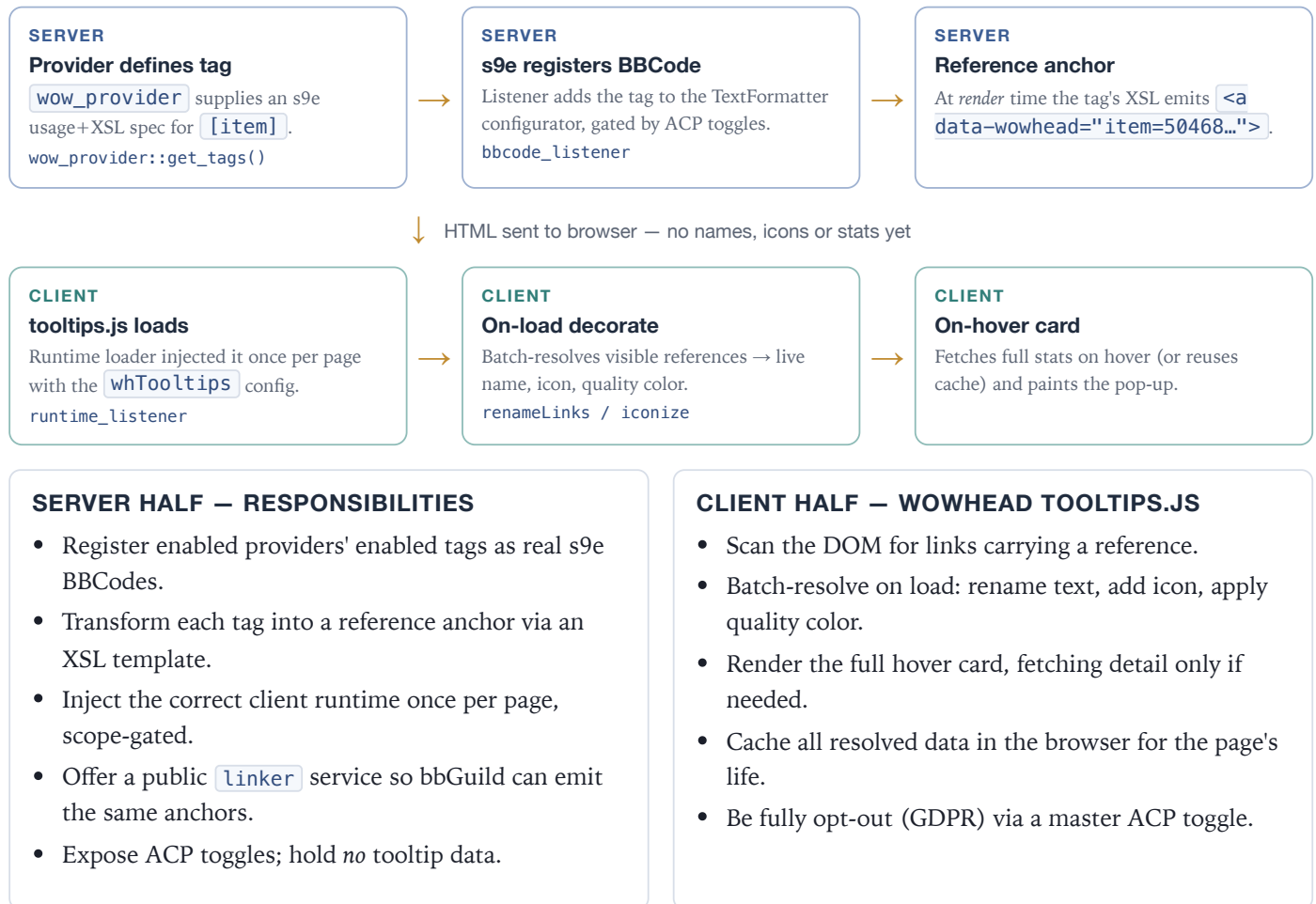
Concern	1.x (MOD)	2.0 (extension)
Platform	phpBB 3.0, PHP 5.x	phpBB 3.3, PHP 8.1+
Tag engine	hand-rolled regex parser	<code>s9e\TextFormatter</code> <code>addCustom()</code>
Data path	server → wowhead.com (per miss)	browser → wowhead.com (async, cached)
Server holds data?	yes (DB cache)	no — reference only
Lookup key	name or numeric ID	numeric ID only — name/text tags dropped
Authoring	type a readable name inline	look the numeric ID up first
Old name-based posts	render tooltips	no longer render (degrade to text/plain link)
Games	WoW only	WoW + Diablo 4 (pluggable)
Failure surface	scraping breaks on WowHead redesigns	none server-side; third-party script is opt-out

Hold onto one idea and the rest of the extension reads easily: the server's entire job is to emit a link WowHead's script will recognize. The provider registry, the `s9e` listener, the runtime loader — they all exist to produce that *one bare reference anchor* and get the script onto the page.

3 The reference-and-enrich model

The extension has two halves — a **server side that emits references** and a **client side that enriches them** — and they meet at exactly one place: an `<a>` tag whose `data-wowhead` attribute (or, for Diablo 4, its `href`) names an entity. That's the whole interface. The server never learns the entity's name, its icon, or a single stat; the browser is the only side that does.

☐ Server (PHP, render time — reference only) ☐ Client (WowHead tooltips.js — enrichment)



Enrichment happens in the browser, talking straight to WowHead, so none of that data ever passes through phpBB. bbTips stores nothing, caches nothing, and makes no network calls of its own — which is precisely why the old scrape-and-cache layer could be dropped rather than rebuilt.

4 End-to-end pipeline — a single `[item]`

Here is one tag, `[item domain=wotlk ench=3825 gems=40133]50468[/item]`, traced from configuration all the way to the pop-up. Stages 0–5 run in PHP; from stage 6 on, it's all the browser.

#	Stage	What happens	Where
0	Install	Migration seeds config (provider/tag toggles, domain, scope, display flags) and the ACP module.	migrations/basics/data.php
1	Configure s9e	On formatter setup, the listener walks enabled providers → enabled tags and calls <code>addCustom(usage, template)</code> .	bbcode_listener · core.text_formatter_s9e_configure_after
2	Parse (store)	When the post is saved, s9e validates the tag against its usage and stores a compact XML representation of it — <i>not</i> the anchor. The XSL template is not run yet.	s9e TextFormatter (parse)
3	Render (emit)	Each time the post is displayed, s9e runs the tag's XSL rendering template → <code>50468</code> .	wow_provider::tag_spec() XSL (render)
4	Page render	On page header, the runtime loader checks scope + master toggle, collects enabled providers' assets, de-dupes by URL.	runtime_listener · core.page_header
5	Inject runtime	Template vars emit the <code>whTooltips</code> config + one <code><script src=tooltips.js></code> into <code><head></code> .	overall_header_head_append.html
6	Script loads	WowHead's <code>tooltips.js</code> reads <code>whTooltips</code> and scans the DOM for reference links.	tooltips.js (browser)
7	On-load decorate	Batch lookup to WowHead resolves name/icon/quality; links are renamed, iconized, colored in place.	wowhead.com data API
8	On-hover fetch	Hovering a link fetches the full tooltip (stats, description) if not already cached.	wowhead.com data API
9	Pop-up painted	The floating card renders at the cursor; results cached client-side for the page.	tooltips.js (browser)

The server touches the tag twice — it stores a representation when the post is saved, then emits the anchor when the post is shown — and after that it's out of the picture. The browser handles everything downstream in two passes: a quick batch *decorate* as the page loads, and a detailed *card* on hover. Both talk to WowHead directly and cache as they go.

Switch off the master runtime toggle, or block the third-party script at the network, and stages 6–9 just don't run. The anchors are still there and still work — they're ordinary links to wowhead.com. The server's output is perfectly valid with or without the script on top of it.

5 Component & service map

Everything is a small, namespaced service under `avathar\bbtips`, wired up in `config/services.yml`. The pure-logic pieces — the provider registry, both providers, `tag_gate`, `linker` and `toolbar_builder` — have PHPUnit tests, and there's a system test around the ext class. The event listeners, the ACP, the migration, and the actual s9e parse/render path don't have tests yet; §10 comes back to that.

```
provider/ # — THE EXTENSIBILITY SEAM — tooltip_provider_interface.php # contract:
get_tags/build_link/get_runtime_asset... abstract_wowhead_provider.php # shared WowHead runtime + anchor builder
wow_provider.php # WoW tags, XSL templates, data-wowhead params diablo4_provider.php # D4 tags →
wowhead.com/diablo-4/... links provider_registry.php # collects tagged providers; get/all/enabled event/ # —
phpBB EVENT LISTENERS — bbcode_listener.php # registers BBcodes with s9e configurator runtime_listener.php #
injects tooltips.js once per page (scoped) posting_listener.php # adds the editor toolbar helper (UI only)
service/ # — PLAIN SERVICES — tag_gate.php # single source of truth: provider/tag enabled? linker.php #
public API for bbGuild (build anchors) toolbar_builder.php # headline buttons + grouped dropdown data acp/
main_module.php · main_info.php # settings page styles/all/template/event/ # head-append runtime + posting
toolbar markup migrations/basics/ data.php # config seed + ACP module registration config/ services.yml ·
language/{en,de,fr,nl} · tests/ · ext.php · composer.json
```

The DI graph

Service	Depends on	Role
provider.wow / .diablo4	config (wow only)	Tagged <code>bbtips.tooltip_provider</code> ; define tags, links, runtime asset.
provider_collection	service_container	phpBB <code>service_collection</code> gathering all tagged providers.
provider_registry	provider_collection	Index by id; <code>get()/all()/enabled()</code> .
tag_gate	config	Are a provider / tag enabled? Shared truth for registration + toolbar.
bbcode_listener	registry, tag_gate	Registers enabled tags with s9e.
runtime_listener	registry, config, template, user	Injects the client runtime, scope-gated.
toolbar_builder	registry, tag_gate, language	Builds editor-helper button/dropdown data.
posting_listener	toolbar_builder, template, language	Feeds the toolbar into the posting editor.
linker	registry	Public façade for other extensions (bbGuild).

The shape is worth noticing: **almost everything leans on the registry**, and the registry leans on nothing but the providers handed to it. That's the joint the whole design pivots on.

The advertised minimum and the real one don't match. Both `composer.json` (`soft-require phpbb/phpbb >=3.3.0`) and `ext.php` (`MIN_PHPBB_VERSION = '3.3.0'`) claim 3.3.0, and `is_enableable()` only ever checks that number. The install migration, though, declares `depends_on() = ['\phpbb\db\migration\data\v33x\v3311']` — so the migrator won't run without 3.3.11. On anything from 3.3.0 to 3.3.10 the enable check passes, and then the migration fails when it can't resolve that dependency, leaving a half-enabled extension. Two ways to square it: raise the declared minimum to 3.3.11 (composer, ext.php, README, and this doc), or point the migration at a dependency that already exists in 3.3.0. Either way, the honest minimum right now is 3.3.11.

6 The provider abstraction

A *provider* is where all the game-specific knowledge lives: which tags exist, how each one turns into a link, and which client runtime the game needs. The core never sees past the interface.

```
interface tooltip_provider_interface {
    function get_id(): string;           // 'wow', 'diablo4'
    function get_game_name(): string;    // ACP label
    function get_tags(): array;          // s9e usage + XSL template per tag
    function get_primary_tags(): array;  // which tags get headline toolbar buttons
    function build_link($type, $id, $opts): string; // anchor for the linker API
    function get_runtime_asset(): array; // { url, config } for the loader
}
```

6.1 Shared base, thin subclasses

The two providers that ship today both extend `abstract_wowhead_provider`. It holds the one thing they have in common — the WowHead runtime — plus a small helper for building a `data-wowhead` anchor:

```
abstract class abstract_wowhead_provider implements tooltip_provider_interface {
    const RUNTIME_URL = 'https://wow.zamimg.com/js/tooltips.js';
    function get_runtime_asset(): array {
        return ['url' => self::RUNTIME_URL,
            'config' => ['colorLinks'=>true, 'iconizeLinks'=>true, 'renameLinks'=>true]];
    }
}
```

Since both hand back the *same* URL, enabling WoW and Diablo 4 together still gets you a single `tooltips.js` — the loader de-dupees them. Diablo 4 simply rides the same script through its `wowhead.com/diablo-4/...` links.

6.2 The registry & the tagged-service seam

Providers get tagged in YAML and gathered up by phpBB's `service_collection` — Symfony's `!tagged_iterator` isn't available in 3.3, so this is the phpBB-native way to do it. The registry then indexes them by id and filters against the ACP toggles:

SERVICES.YML

```
avathar.bbtips.provider.wow:
  class: ...\wow_provider
  arguments: [ '@config' ]
  tags:
    - { name: bbtips.tooltip_provider }

avathar.bbtips.provider_collection:
  class: phpbb\di\service_collection
  arguments: [ '@service_container' ]
  tags:
    - { name: service_collection,
        tag: bbtips.tooltip_provider }
```

PROVIDER_REGISTRY.PHP

```
function __construct(iterable $providers){
    foreach($providers as $p)
        $this->providers[$p->get_id()] = $p;
}

function enabled(array $ids): array {
    return array_intersect_key(
        $this->providers, array_flip($ids));
}
```

Adding Guild Wars 2 later: the seam gets you part of the way, not all of it. The registry itself is generic, so a provider class plus one tagged `services.yml` line is enough for three things to notice the new game on their own — s9e BBCode registration (`bbcode_listener` walks `registry->all()`), the editor toolbar (`toolbar_builder`), and runtime de-duplication. The rest still names its games by hand, and each of these would need a touch:

- `runtime_listener::enabled_provider_ids()` spells out `wow` and `diablo4` — a third provider's runtime simply won't load until it's listed here.

- The migration (`data.php`) hardcodes the `bbtips_provider_*` and `bbtips_tag_*` config seeds.

- The ACP module (`main_module.php`) carries the provider/tag key lists, and its template (`acp_bbtips_settings.html`) has a fieldset per provider.
- The language files need labels and help strings for the new tags.

So a realistic GW2 estimate is: a provider class and a service tag, plus edits to the runtime listener, the migration, the ACP module and its template, and the language files. The discovery seam earns its keep — but "add a class and you're done" oversells it.

7 BBCode registration via s9e TextFormatter

phpBB 3.3 swapped the old regex BBCode system for the s9e\TextFormatter engine. A custom tag with named attributes can't be dropped into a `bbcode.html` file anymore; it has to be registered through a configurator event. That's all `bbcode_listener` does:

```
function getSubscribedEvents(){
    return ['core.text_formatter_s9e_configure_after' => 'register_bbcodes'];
}

function register_bbcodes($event){
    $configurator = $event['configurator'];
    foreach($this->registry->all() as $id => $provider){
        if(!$this->gate->provider_enabled($id)) continue; // ACP provider toggle
        foreach($provider->get_tags() as $tag){
            if(!$this->gate->tag_enabled($tag['bbcode'])) continue; // ACP per-tag toggle
            try { $configurator->BBCodes->addCustom($tag['usage'], $tag['template']); }
            catch(\Exception $e) { /* a bad spec must never break the whole board */ }
        }
    }
}
```

7.1 The usage string and the XSL template

Every tag comes as a pair. The usage string declares the attributes and the content type — s9e uses it to validate the tag and parse it into stored XML. The rendering template is an XSL fragment that s9e runs later, *at render time*, to turn that stored XML into the anchor. The WoW provider generates both, per tag:

```
// usage: numeric content + optional named attributes
[ITEM domain={SIMPLETEXT1;optional} ench={UINT;optional}
    gems={REGEXP=/^\d+(:\d+)*$/;optional} pcs={REGEXP=/^\d+(:\d+)*$/;optional}] {UINT2} [/ITEM]

// template (simplified): href + a data-wowhead attribute assembled with XSL conditionals
<a href="https://www.wowhead.com/item={@content}">
  <xsl:attribute name="data-wowhead">
    item=<xsl:value-of select="@content"/>
    <xsl:if test="@domain">&domain=<xsl:value-of select="@domain"/></xsl:if>
    <xsl:if test="@ench">&ench=<xsl:value-of select="@ench"/></xsl:if>
    <xsl:if test="@gems">&gems=<xsl:value-of select="@gems"/></xsl:if>
  </xsl:attribute><xsl:apply-templates/></a>
```

Mechanism	Purpose
Type tokens (<code>UINT</code> , <code>SIMPLETEXT</code> , <code>REGEXP</code>)	s9e validates attributes at parse time — <code>gems</code> / <code>pcs</code> accept only colon-separated numbers; content must be an unsigned int.
<code>domain</code> XSL choose	A per-tag <code>domain=</code> wins; otherwise the board-wide default (<code>bbtips_wow_domain</code>) is baked into the template. Empty values are omitted entirely.
<code>itemico</code> size	Adds a <code>data-wh-icon-size</code> attribute (<code>small/medium/large</code>) instead of a separate entity type.
Diablo 4 tags	Simpler: <code>[d4item]{UINT}[/d4item]</code> → <code></code> ; no <code>data-wowhead</code> needed — the D4 script decorates from the href.

Two small safety measures are worth calling out: the listener skips any tag already on the configurator, and it wraps `addCustom()` in a try/catch. One malformed spec, then, can't take post rendering down across the whole board.

There's a subtlety here that the parse-vs-render split makes concrete. The usage and template are built from config — the default `bbtips_wow_domain`, for instance, is written straight *into* the XSL — and compiled once into phpBB's `text_formatter.cache`. Since the anchor is produced at render time from that cached template rather than frozen when the post was parsed, changing the default domain in the ACP has to invalidate the cache to take effect — which is why the save

handler calls `text_formatter.cache->invalidate()`. Re-render an old post afterward and it quietly picks up the new domain and tag toggles.

8 Runtime loader & client enrichment

8.1 Injecting the script once

`runtime_listener` hangs off `core.page_header`. If the master toggle is off or the page is out of scope, it does nothing. Otherwise it gathers the runtime asset from each enabled provider, **de-dupes by URL**, lays the admin's display options over whatever the providers default to, and passes the result to the template:

```
function inject_runtime($event){
    if(!(int)$this->config['bbtips_runtime_enabled'] || !$this->in_scope()) return;
    $srcs = []; $cfg = [];
    foreach($this->registry->enabled($this->enabled_provider_ids()) as $provider){
        $asset = $provider->get_runtime_asset();
        $srcs[$asset['url']] = true; // ← de-dupe: wow + diablo4 share one URL
        $cfg = array_merge($cfg, $asset['config']);
    }
    $cfg['colorLinks'] = (bool)(int)$this->config['bbtips_color_links']; // ACP overrides
    $cfg['iconizeLinks'] = (bool)(int)$this->config['bbtips_iconize_links'];
    $cfg['renameLinks'] = (bool)(int)$this->config['bbtips_rename_links'];
    $this->template->assign_vars(['S_BBTIPS_RUNTIME'=>true,
        'BBTIPS_WHTOOLTIPS_JSON'=>json_encode($cfg)]);
    foreach(array_keys($srcs) as $src)
        $this->template->assign_block_vars('bbtips_runtime_srcs', ['SRC'=>$src]);
}
```

The template event `overall_header_head_append.html` renders it into the page head:

```
<!-- IF S_BBTIPS_RUNTIME -->
<script>const whToolTips = {BBTIPS_WHTOOLTIPS_JSON};</script>
<!-- BEGIN bbtips_runtime_srcs -->
<script src="{bbtips_runtime_srcs.SRC}"></script>
<!-- END ... / ENDIF -->
```

8.2 Scope & the master toggle

Config	Effect on the client half
bbtips_runtime_enabled	Master GDPR switch. Off ⇒ no third-party script at all; reference anchors stay as plain links.
bbtips_scope	<code>all</code> (default) loads on every page incl. <code>app.php</code> /bbGuild; <code>posts</code> restricts to <code>viewtopic.php</code> / <code>posting.php</code> .
bbtips_color / iconize / rename_links	Feed the <code>whToolTips</code> object — quality colors, inline icons, live-name replacement.

8.3 What tooltips.js does in the browser

Once it loads, WowHead's script reads `whToolTips` and goes to work in two async passes. Neither touches phpBB:

PASS A · ON LOAD

Decorate

Scans reference links, batch-resolves ids → name, icon, quality; applies

`renameLinks` /icon/color in place.



PASS B · ON HOVER

Full card

On hover, fetches full stats/description (or reuses cache) and renders the floating tooltip at the cursor.



CLIENT CACHE

Reuse

All resolved data is cached in the browser for the page's lifetime; repeat hovers are instant.

9 Editor helper, linker & configuration

9.1 The posting-editor helper (UI only)

Nobody wants to type brackets from memory, so `posting_listener` — on `core.posting_modify_bbcode_status`, and only when the forum actually allows BBCode — asks `toolbar_builder` which tags are enabled and drops a toolbar block into the editor. The builder takes each provider's `get_primary_tags()`, promotes up to two of them to **headline buttons**, and tucks the rest into a per-game **dropdown**:

TOOLBAR_BUILDER::BUILD()

- Walks enabled providers → enabled tags via the same `tag_gate` as registration.
- Headline = first `MAX_HEADLINE` (2) primary tags (WoW: `item`, `spell`).
- Remaining tags grouped by `get_game_name()` for an `<optgroup>` dropdown.
- Returns `{headline, groups, has_any}` — pure data, no markup.

WHAT THE AUTHOR SEES

- `[item]` / `[spell]` buttons insert an empty tag skeleton with the cursor between the tags.
- A "More tooltips..." dropdown lists the rest, grouped WoW / Diablo 4, with a help line.
- A hint under "BBCode is ON" makes the tags discoverable.
- Works even with the tooltip script disabled — it only inserts text.

The markup lives in `posting_editor_buttons_after.html` + `posting_editor_bbcode_status_after.html`; a tiny inline `bbtips_insert()` JS handles the dropdown. Colors are inherited/opacity-softened so it fits light (prosilver) and dark (pbwow3/pbtech) styles without a hardcoded palette.

9.2 The linker service — bbGuild bridge

The public `avathar.bbtips.linker` service is how another extension can emit a correct anchor without knowing the first thing about providers — it just delegates to the provider's `build_link()`:

```
$linker->wow()->build_link('item', 50468, ['ench'=>3825, 'gems'=>[40133], 'domain'=>'classic']);  
// → <a href=... data-wowhead="item=50468&domain=classic&ench=3825&gems=40133">50468</a>
```

This is how bbGuild reuses bbTips without being tied to it. On a character page, bbGuild renders equipped gear through the linker if `avathar.bbtips.linker` is in the container, and falls back to a plain link if it isn't. The check is just "does the service exist" — no composer dependency either way — and because `scope=all` already put the runtime on that page, the gear links light up like any other.

9.3 ACP & migration

One ACP page (`acp/main_module.php`, under `ACP_CAT_BBTIPS`) covers all the knobs: provider enables, per-tag enables, the default WoW domain, the display options, scope, and the master runtime toggle. A single squashed migration (`migrations/basics/data.php`) seeds every one of those config keys — tags default *on* — and registers the ACP category and module. `effectively_installed()` keys off `bbtips_scope`.

10 Design observations

STRENGTHS

- **Deleted a whole failure class.** No server scraping/caching means no breakage when WowHead redesigns, and no stale cache.
- **One seam, many games.** The provider interface + tagged registry make a new game a self-contained add-on.
- **Config-gated everywhere.** A single `tag_gate` drives registration *and* the toolbar, so they can never disagree.
- **Testable — and partly tested.** The pure-logic services have PHPUnit coverage under CI (a different world from the MOD's includes); the listeners, ACP, migration and s9e parse/render are not yet covered and the functional suite is empty — a clear next step, not a claim of full coverage.
- **Honest privacy story.** The third-party script is a documented, one-click opt-out that degrades to plain links.

TRADE-OFFS & CONSTRAINTS

- **Lost capability: name/text tags.** The MOD's by-name tags are gone; authors must supply numeric IDs and all name-based back-content stops rendering tooltips. The single clearest regression from 1.0.7.
- **Third-party client dependency.** Tooltips need WowHead's CDN reachable in the reader's browser.
- **Reader-side data cost.** Enrichment traffic moves to each visitor rather than being cached once server-side.
- **Classic-era attributes.** `gems` / `ench` / `pcs` are meaningful mainly against classic domains — hence the load-bearing `domain` switch.
- **GW2 still pending.** The discovery seam is ready, but enablement config, ACP, migration and the runtime listener are hardcoded per game, so a new provider is not purely additive (§6).
- **Version metadata mismatch.** Declared minimum is 3.3.0, but the install migration needs 3.3.11 — align them before treating 3.3.0 as supported (§5).

If you take one thing away: bbTips 2.0 registers game BBCodes with s9e, renders each to a bare reference anchor, and lets WowHead's `tooltips.js` — loaded once per page — turn those references into live pop-ups in the reader's browser. Nothing is fetched, cached, or stored server-side. The provider layer is what keeps that machinery open to new games.

Appendix — configuration keys

Key	Default	Meaning
<code>bbtips_provider_wow / _diablo4</code>	1 / 1	Enable a provider's tags + runtime.
<code>bbtips_tag_<bbcode></code>	1	Per-tag on/off (item, itemico, spell, quest, craft, achievement, itemset, npc, d4item, d4skill).
<code>bbtips_wow_domain</code>	" (retail)	Default WowHead domain when a tag omits <code>domain=</code> .
<code>bbtips_scope</code>	all	Where the runtime loads: <code>all</code> or <code>posts</code> .
<code>bbtips_runtime_enabled</code>	1	Master third-party-script switch (GDPR opt-out).
<code>bbtips_color / iconize / rename_links</code>	1 / 1 / 1	WowHead display behaviors passed via <code>whTooltips</code> .